

DS&AI Additional Admissions Form



Academic year 2026-2027

1. Applicant

Full name	John Smith
TU/e student number	1234567

2. Previous studies

Degree programme	Bachelor Computer Science and Engineering
Graduation date	12/08/2026
University	Eindhoven University of Technology
Location	Eindhoven, The Netherlands
Full-time equivalent duration	3 years
Credit system	ECTS
Total degree credits	180
Grading system	Numeric - higher grades are better
Best possible grade	10
Minimum passing grade	6.0

3. Prerequisite coverage

3.1 Calculus

Required topics

- Continuity
- Closed, open, and bounded sets
- Differentiation, integration
- Distances, norms

Evidence

No relevant course selected.

3.2 Linear Algebra

Required topics

- Calculate with matrices and vectors

- Solve linear systems with Gauss(-Jordan) elimination
- Understand and apply rank, orthogonality, (in)dependency, eigenvalue decomposition and eigenvectors

Evidence

Course code	Course title	EC	Grade
2DRR00	Linear Algebra and Applications	5	8
Degree	Degree 1: Bachelor Computer Science and Engineering (Eindhoven University of Technology)		
Official course description	Content Informal description This course offers a wide range of interesting linear algebra techniques and very nice applications, including Big Data applications. These techniques are almost indispensable for the current Big Data era, and are also relevant for the field of Artificial Intelligence, including Deep Learning. You will learn how to solve linear systems, arguably the most common and important scientific problem. Least squares methods (or regression, very useful in Engineering) can be used to determine an approximate line (or polynomial, or spline) through a number of points (used in fitting problems, Computer Graphics, games, movies). Rotations and reflections may be useful for Computer Graphics and Robotics. Angles between vectors can be used to compare tastes of movies, music, or books, and to predict such tastes for the future. You will learn the idea behind Google PageRank and its connection with eigenvalues, how it is computed, and you can also apply similar techniques to finally determine which team or country is the best soccer team. We can use text mining methods to determine the most important tweets or keywords from a set of tweets or similar data from social media; useful for effective communication. Similar data mining techniques may be used to study customer behavior. The course is designed to be interesting for many students, including (but not restricted to!) students from Computer Science, Data Science, P&T, and Robotics. The subjects are fascinating on their own, but also form an ideal preparation for subsequent courses such as Data Mining, Machine Learning, Numerical Linear Algebra, Computer Graphics, and Optimization. The course offers a nice combination of theoretical (math) and practical (implementation) work, partly done in groups. You can use a language you might already know (for instance Python, Matlab, R) or learn a new language (for instance Julia). Content Matrices, vectors • Operations, matrix and vector norms, angles between vectors • Applications in recommender systems Linear systems and least squares problems • Gaussian elimination, Gauss and Gauss-Jordan form, solutions, rref, inverse matrices • Normal equations		

	• Applications in data fitting Rotations, projections, and reflections • Corresponding matrices • Applications in Computer Graphics and robotics Eigenvalues and eigenvectors • Eigenvalue decomposition • Determinants • Google PageRank, clustering, and other applications in Markov chains Singular value decomposition (SVD/PCA) • Singular values and vectors • Orthogonality, rank, linear dependency, low-rank matrices, TSVD • Applications in Web Analytics and data/text mining, feature extraction Implementing linear algebra • Flops and memory costs, finite precision arithmetic and rounding errors • Linear algebra with Python, Julia, Matlab, or another language of choice
Additional explanation	-
Also used for	-

3.3 Probability and Statistics

Required topics

- Introductory probability theory (single-variable & multi-variable distributions, marginalization, conditioning, independence, Bayes' rule)
- Knowledge of discrete and continuous random variables
- Descriptive statistics including the theory and practice of confidence intervals, hypothesis testing and estimation theory

Evidence

No relevant course selected.

3.4 Discrete Mathematics

Required topics

- Propositional logic and predicate logic
- Reason with logical formulas
- Standard proving techniques such as induction, case distinction, contradiction
- Set operations, functions, relations, orderings
- Knowledge of basic graph structures (graphs, trees, and their properties)

Evidence

No relevant course selected.

3.5 Data Structures and Algorithms

Required topics

- Use of standard data structures
- Algorithm design techniques
- Reason about algorithm complexity and efficiency

Evidence

Course code	Course title	EC	Grade
2IL50	Data Structures	5	7.5
Degree	Degree 1: Bachelor Computer Science and Engineering (Eindhoven University of Technology)		
Official course description	Objectives Learning goals: There are many aspects to the study of data structures, and the algorithms that operate upon them. In this course, the student will learn the basic skills and knowledge to develop efficient algorithms to solve computational problems and to make informed choices between different solutions for the same problem. Learning Outcomes: After completing the course, you have a working knowledge of fundamental data structures: heaps, hash tables, binary search trees, and union-find; basic sorting algorithms such as InsertionSort, MergeSort, and QuickSort; basic graph algorithms such as minimum spanning trees, breadth- and depth-first search, and topological sorting. Furthermore, you can apply basic algorithm design techniques (incremental, recursive, divide-and-conquer); use standard proof techniques (induction, invariants) to establish the correctness of algorithms; characterize the efficiency of algorithms by performing a worst-case asymptotic analysis use the skills above to make informed choices between different solutions for the same computational problem. Content Design techniques: Incremental algorithms, recursion, divide & conquer. Correctness: induction and invariants. Efficiency analysis: O-notation, recurrences. Sorting: MergeSort, InsertionSort, HeapSort, sorting in linear time, lower bounds for sorting. Selection Algorithms. Data structures: abstract data structures, heaps, hash tables, search trees (incl. red-black trees), augmenting data structures, union-find. Basic graph algorithms: adjacency list, adjacency matrix, DFS, BFS, topological sort, minimum spanning trees.		
Additional explanation	In the assignments we covered 'Reason about algorithm complexity and efficiency'.		
Also used for	-		

3.6 Programming and Software Development

Required topics

- Write programs from scratch in imperative and object-oriented languages (not only scripts)
- Use general algorithmic techniques (aggregation, searching, sorting, recursion)
- Apply the principles of code quality and software engineering

Evidence

No relevant course selected.

3.7 Data Modeling and Databases

Required topics

- Design data models (E-R diagrams, UML) from natural language requirements
- Query relational databases (SQL) based on natural language requirements

Evidence

No relevant course selected.

3.8 Machine Learning and Data Mining

Required topics

- Apply feature selection and extraction
- Apply supervised learning (classification and regression)
- Apply unsupervised learning (clustering and matrix factorization)
- Apply evaluation methods and understand overfitting
- Training machine learning models and making predictions

Evidence

No relevant course selected.

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Output schema version: 1

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